

Student Information

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Submitted: 2026-07-23 06:33:20

SCORE: 3/28 (10.7%) - Grade F

Section Breakdown

Section	Score	Total	Pct
K-Means Algorithm	0	5	0%
Mean Shift Algorithm	0	3	0%
DBSCAN Algorithm	0	3	0%
Gaussian Mixture Model (GMM)	1	4	25%
Real-World Data Preprocessing	0	3	0%
Visualization Utilities	1	4	25%
Evaluation Metrics	0	2	0%
Model Selection and Interpretation	1	4	25%

Questions to Review

Q1 (Correct: B)

What is the primary objective function that K-Means clustering aims to minimize?

K-Means
assignment
Objective

Q2 (Correct: B)

How does K-Means++ initialization improve upon random initialization in K-Means clustering?

K-Means++
convergence
number

Q3 (Correct: C)

What is a key limitation of K-Means when applied to customer segmentation regarding cluster shapes?

K-Means
assigns
well to

Q4 (Correct: B)

In the K-Means algorithm, what happens during the assignment step?

During
update
GMM

Q5 (Correct: C)

Which scenario would be most problematic for K-Means clustering?

K-Means
ones to
handle

Q6 (Correct: B)

What key parameter does Mean Shift not require as input compared to K-Means?

Mean
Option
conver

Q7 (Correct: B)

How does the bandwidth parameter influence clustering granularity in Mean Shift?

Bandw
produc
and D

Q8 (Correct: B)

How does Mean Shift handle clusters with arbitrary shapes compared to K-Means?

Mean
discov
prepro

Q9 (Correct: C)

DBSCAN classifies data points into three types. Which of the following is NOT one of them?

DBSCAN
sufficie
neither

Q10 (Correct: B)

What happens if epsilon is set too small in DBSCAN?

If epsilon
points
incore

Q11 (Correct: B)

What unique capability does DBSCAN offer for customer segmentation that K-Means does not?

DBSCAN
errors.
DBSCAN

Q12 (Correct: B)

How does GMM fundamentally differ from K-Means in its assignment of data points to clusters?

GMM
one cl
algorith

Q13 (Correct: B)

What is the purpose of the E-step in GMM's Expectation-Maximization algorithm?

The E-
of men
not the

Q15 (Correct: B)

What does a lower BIC value generally indicate when comparing GMM models?

A lowe
opposi

Q16 (Correct: C)

What is the purpose of StandardScaler in clustering preprocessing?

Stand
domin
metho

Q17 (Correct: B)

Why is feature scaling important before applying K-Means clustering?

K-Mea
assign
about

Q18 (Correct: B)

Which preprocessing step would be most appropriate for handling missing values in customer data before clustering?

Mean
Option

Q20 (Correct: B)

Which visualization is most useful for determining the optimal number of clusters in K-Means?

The el
returns
Option

Q21 (Correct: B)

What does a Silhouette plot visualize for a clustering solution?

A Silh
cluster

Q22 (Correct: B)

When visualizing clustering results in 2D, which technique is commonly used for dimensionality reduction?

PCA r
results

Q23 (Correct: B)

For Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Score, which combination indicates better clustering?

Better

Q24 (Correct: B)

What does a Silhouette Score near +1 indicate about a customer's cluster assignment?

A Silh
strong
yield a

Q25 (Correct: C)

Which algorithm is most appropriate when you suspect clusters have different shapes and densities, and you want to identify outliers?

DBSC
K-Mea
cluster

Q26 (Correct: C)

You have customer data with overlapping segments where customers may belong to multiple segments with different probabilities. Which algorithm is best suited?

GMM
memb

Q27 (Correct: B)

When comparing clustering results from different algorithms on the same dataset, which approach is most appropriate?

Proper
qualita